

Using the SCC

MedAI Hackathon

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- We're here to help you use the SCC for the Hackathon.



- Do not use the “BU Guest” WiFi network.
- The browser access to the SCC will fail in numerous ways.
- Use the [eduroam](#) WiFi network
 - Documentation on connecting is [here](#).

SCC Access

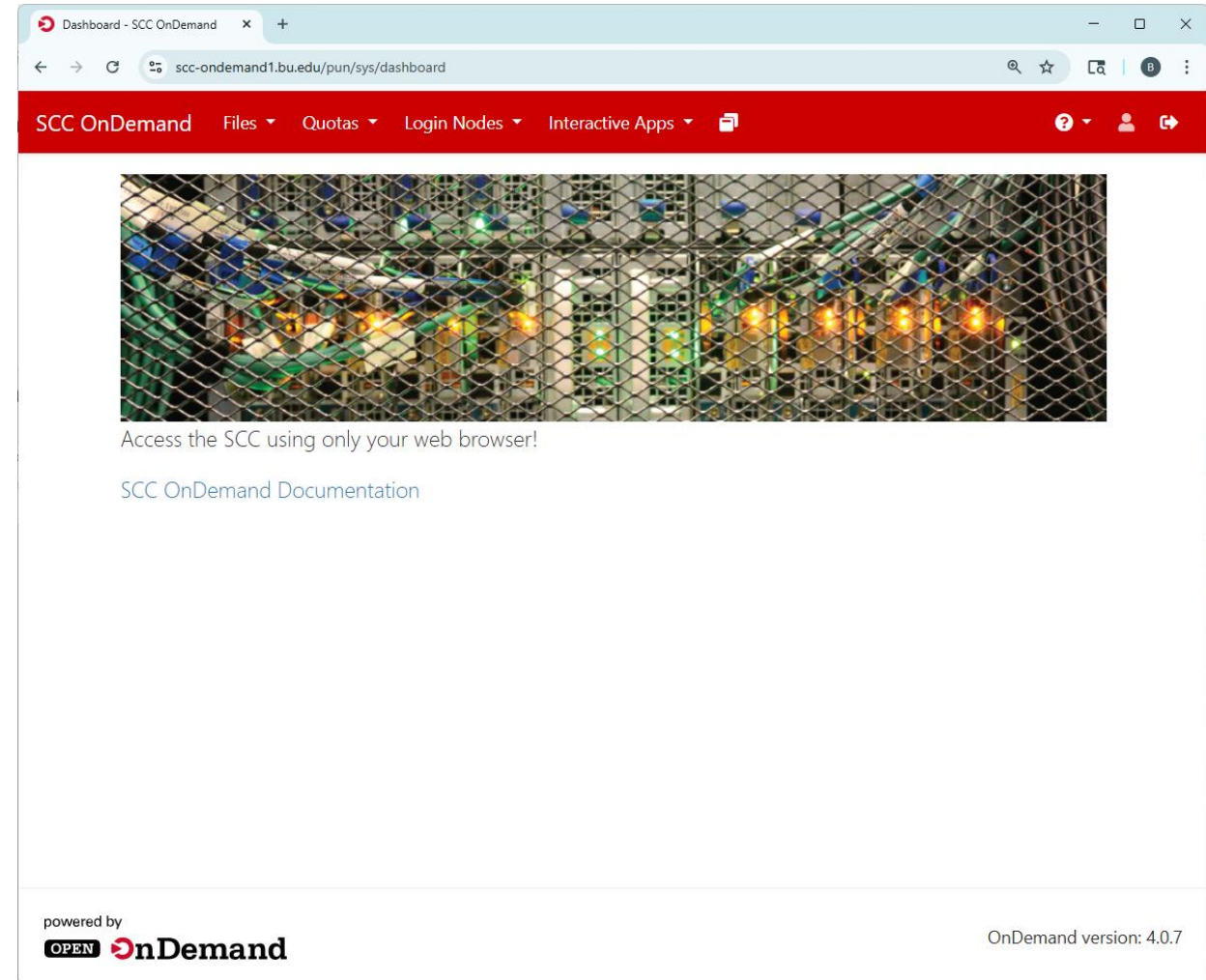
- The Hackathon will use our OnDemand site to access the SCC. Open this URL now.

<https://scc-ondemand.bu.edu>

- Google Chrome is the recommended browser
 - This browser properly handles copy & paste in OnDemand Desktops.
 - The 1st time Chrome opens an OnDemand Desktop you will be asked to enable a plugin. Say yes.
- Beyond interactive use:
 - You can create and submit non-interactive “batch” jobs to the cluster job queue.

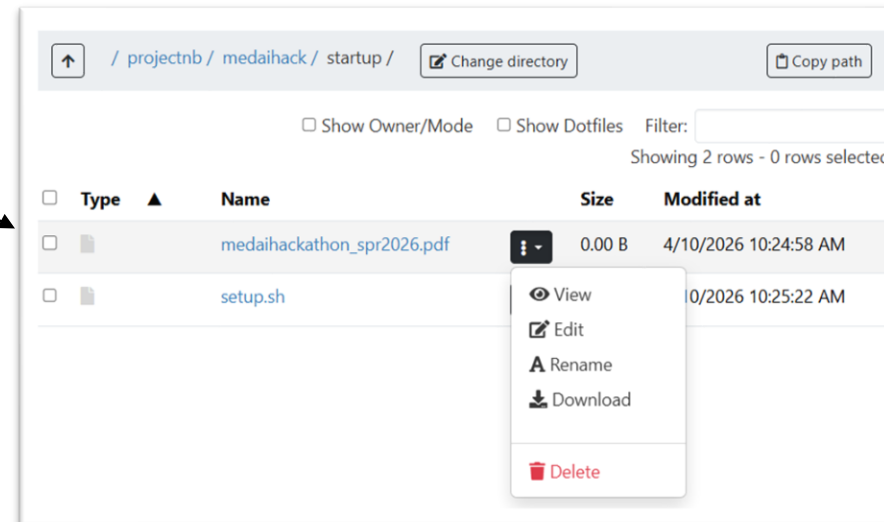
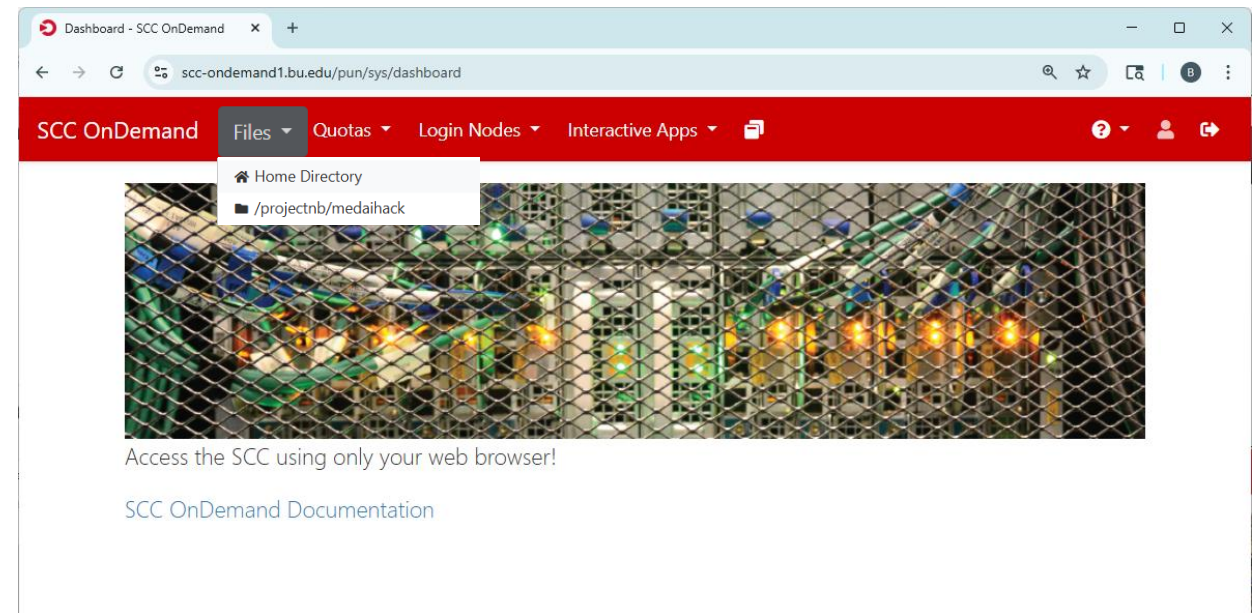
OnDemand Login Screen

- When you have logged in you can click on the menu items.
- “Interactive Apps” is how you will run interactively using Jupyter Notebook or VS Code.



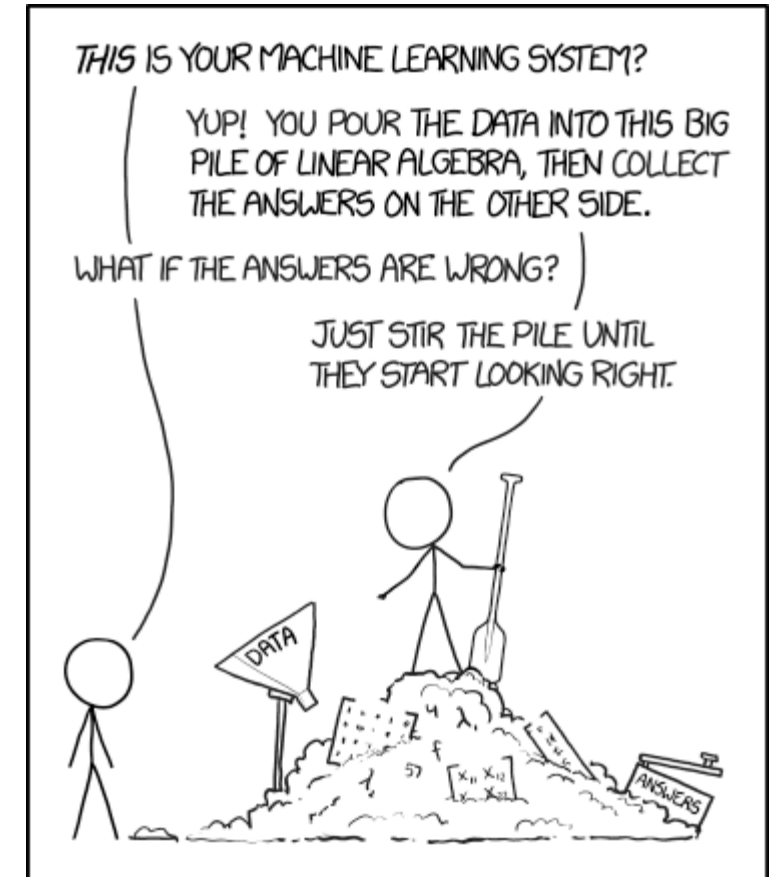
Files

- Click on the Files menu
 - Select `/projectnb/medaihack`
- In the directory listing that shows up click the **startup** directory.
- Click the menu next to *medaihackathon_spr2026.pdf* and download these slides.



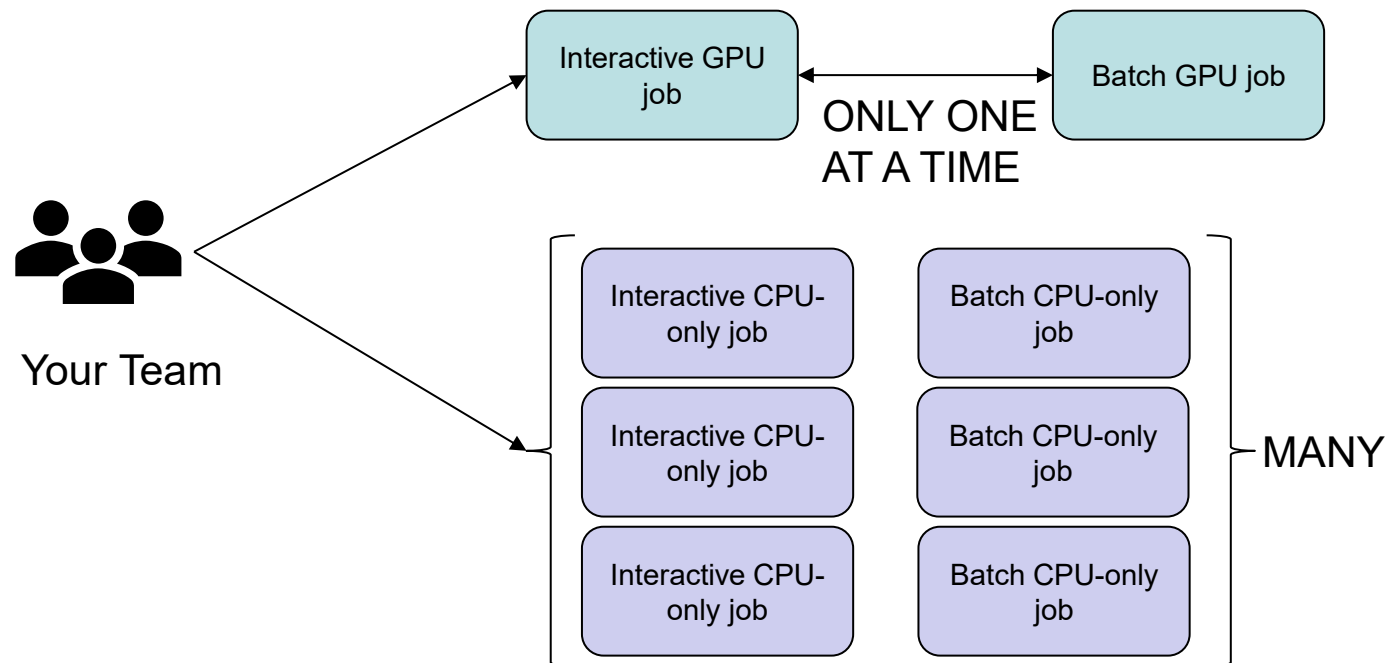
Hackathon Cluster Storage

- The hackathon directory is:
`/projectnb/medaihack`
- This has a storage quota of 20TB.
- Your team has a directory:
`/projectnb/medaihack/teamN`
 - The teams are all sharing the 20TB quota – be mindful of your usage.



Obligatory [XKCD](#) comic

Computing Resources Per Team



Nvidia L40S

48GB of RAM

Ada Lovelace architecture

compute capability: 8.9

76.3 billion transistors

18,176 CUDA parallel processing
cores

568 tensor cores

142 RT cores

- Interactive jobs are ones you're actively using, such as Jupyter Notebooks or VS Code.
- Batch jobs are ones you've submitted to the cluster scheduler to be run without interactivity.

Job Limits

GPU Jobs

- Each team has **1** Nvidia L40S GPU reserved.
- Request **8 CPU** cores and 1 GPU.
- Limit yourself to 64GB of CPU RAM.
- **ONLY ONE** GPU job (interactive OR batch) will **run at a time**.
 - You can queue multiple batch jobs that will use the GPU but **ONLY ONE** will execute at a time.

CPU Jobs

- Teams can run as many CPU-only jobs as they wish.
- These can use multiple cores. We suggest 4 per job unless you know you need more.
- Assume your CPU jobs can use up to 8GB of RAM per requested core.

Hackathon SCC Restrictions



Data Files

- The data files for each challenge are stored in the challenge directories.
- **DO NOT:**
 - Copy them to your team directory
 - Copy them to your own computer
 - Copy them to any other location on the SCC.

Team Directories

- You have full read/write access to your team directory:
 - `/projectnb/medaihack/teamN`
- You do not have access to the directories belonging to other teams

Other SCC resources

- **DO NOT:**
 - Use any GPU that's not part of this hackathon.
 - Use any storage space outside of your team directory:
 - `/projectnb/medaihack/teamN`

Getting SCC help

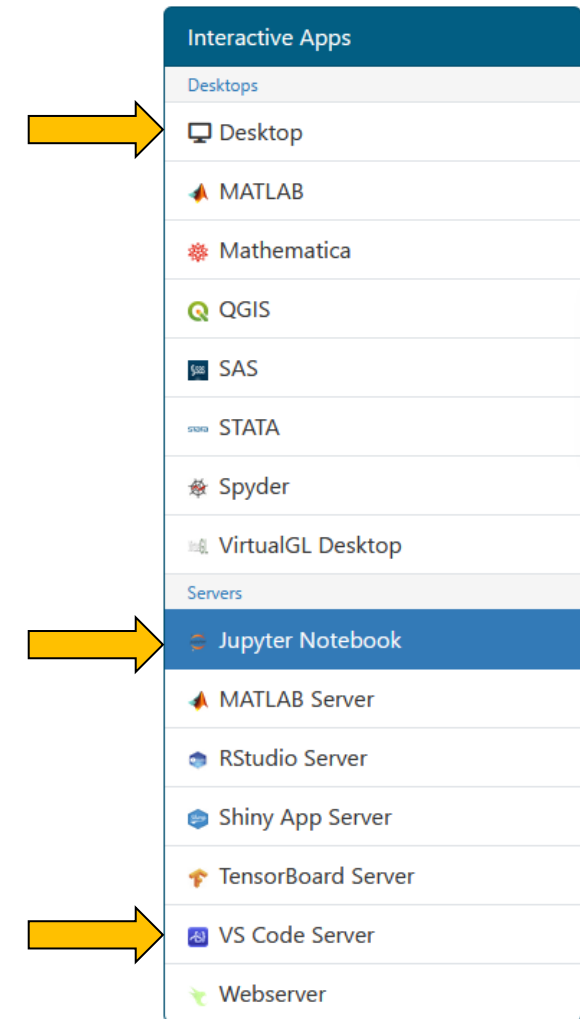
- Brian, Josh, and Atish from Research Computing Services are on site for SCC help.
 - If you have a problem using the SCC, just find us!
- **DO NOT:**
 - Open ServiceNow tickets for the hackathon
 - Email RCS – we won't be monitoring these.

Use project *medaihack*

- All jobs (interactive or batch), *must* specify that they are part of the *medaihack* project.
- More on this in the upcoming slides.

OnDemand Apps

- In OnDemand, click the *Interactive Apps* menu item to see this list.
- To use Python in this Hackathon, we recommend you use these interactive apps:
 - Desktop
 - Visualization, editors other than VS Code
 - Jupyter Notebook
 - Run notebooks using Jupyter Lab or Notebook.
 - Code Server
 - This is to run VS Code in your browser



OnDemand Jupyter

List of modules to load:

```
medaihack/spring-2026 python3/3.12.4
```

Pre-launch command:

```
source /projectnb/medaihack/teamN/venv_name/bin/activate
```

Replace **teamN** and **venv_name** as appropriate.

Working Directory:

```
/projectnb/medaihack/teamN
```

Project:

medaihack

This is **very important!**
Do not use any other project!

Extra qsub option
for GPU jobs:
`-l gpu_type=L40S`

Jupyter Notebook

This app will launch a Jupyter Notebook server on a compute node.

List of modules to load (space separated)

```
medaihack/spring-2026 python3/3.12.4
```

Select Modules

Pre-Launch Command (optional)

```
source /projectnb/medaihack/teamN/venv_name/bin/activate
```

Interface

lab

Working Directory

```
/projectnb/medaihack/teamN
```

Select Directory

The directory to start Jupyter in. (Defaults to home directory.)

Extra Jupyter Arguments (optional)

Number of hours

12

Number of cores

8

Number of gpus

1

GPU compute capability

8.9 (L40S or ...)

Project

medaihack

Extra qsub options

```
-l gpu_type=L40S
```

☐ I would like to receive an email when the session starts

Launch

Choose the Jupyter Lab or
Notebook interface

GPU JOBS

8 cores
1 GPU
8.9 compute capability

CPU JOBS

1, 4, 8, 16, 28, or 32 cores
(as **needed**)
0 GPU

Leave this *empty*
for CPU jobs

OnDemand VS Code Server

List of modules to load:

`medaihack/spring-2026 python3/3.12.4`

Pre-launch command:

```
source /projectnb/medaihack/teamN/venv_name/bin/activate
```

Replace `teamN` and `venv_name` as appropriate.

Working Directory:

`/projectnb/medaihack/teamN`

Project:
medaihack

This is **very important!**
Do not use any other project!

Extra qsub option
for GPU jobs:
`-l gpu_type=L40S`

VS Code Server

This app will launch a [VS Code](#) server using [Code Server](#) on a compute node.

Codeserver Version

4.105.1

Additional modules to load (space separated, optional)

medaihack/spring-2026 python3/3.12.4

Select Modules

Pre-Launch Command (optional)

source /projectnb/medaihack/teamN/venv_name/bin/activate

Working Directory

/projectnb/medaihack/teamN

Select Directory

The directory to start in. (Defaults to home directory.)

Number of hours

12

Number of cores

8

Number of gpus

1

GPU compute capability

8.9 (L40S or ...)

Project

medaihack

Extra qsub options

-l gpu_type=L40S

☐ I would like to receive an email when the session starts

☐ Save settings

Save settings and close

Launch

GPU JOBS

8 cores
1 GPU
8.9 compute capability

CPU JOBS

1, 4, 8, 16, 28, or 32 cores
(as **needed**)
0 GPU

Leave this *empty*
for CPU jobs

OnDemand Desktop

List of modules to load:

`medaihack/spring-2026 python3/3.12.4`

Working Directory:

`/projectnb/medaihack/teamN`

Project:
medaihack

This is ***very important!***
Do not use any other project!

In the terminal window that appears in the Desktop activate your virtual environment:

`source /projectnb/medaihack/teamN/venv_name/bin/activate`

Extra qsub option for
GPU jobs:

`-l gpu_type=L40S`

List of modules to load (space separated)

`medaihack/spring-2026 python3/3.12.4`

Select Modules

Working Directory

`/projectnb/medaihack/teamN`

Select Directory

The directory to start in. (Defaults to home directory.)

Initial command to run

`xfce4-terminal`

Number of hours

12

Number of cores

8

Number of gpus

1

GPU compute capability

8.9 (L40S or ...)

Project

medaihack

Extra qsub options

`-l gpu_type=L40S`

☐ I would like to receive an email when the session starts

☐ Save settings

Save settings and close

Launch

GPU JOBS

8 cores
1 GPU
8.9 compute capability

CPU JOBS

1, 4, 8, 16, 28, or 32 cores
(as **needed**)
0 GPU

Leave this *empty*
for CPU jobs

Saturday Startup

Step 1

- Go to the Files screen on OnDemand and open `/projectnb/medaihack/startup`
- Read the `teams.csv` file to match your team name with the team number.
- You can download this file and open it with something like Excel if that's more convenient.

Step 2

- Start a Desktop session (1 CPU, no GPUs) as shown in the earlier slide.
- Use your **team number** as the working directory:
 - `/projectnb/medaihack/teamN`
- In the terminal window, run this command to retrieve the challenge codes:
 - `/projectnb/medaihack/startup/setup.sh`

Step 3

- Use this Desktop to create the Python virtual environments for the challenges that interest you.
- Once the venv's are created, you can launch Jupyter or VS Code Server jobs that use them.

Sharing the Team GPU

Python Virtual Environments

- You can have more than 1 venv
 - Give them different names.
 - There will be least 1 per challenge that you work on.
 - The venv creator is the **only** person who can add packages to it or modify it.
- Only use virtual environments created with the *virtualenv* command.
 - To avoid hackathon complications, don't use *conda* or *uv*.

1 GPU Job At a Time

- If team member A is using a GPU in a Notebook or in VS Code or is running a batch job, no one else in the team can access it.
- Multiple CPU-only jobs is OK!
 - Team members B, C, and D can all run their own notebooks or Desktop sessions on CPUs at the same time.

Batch Jobs

- If a batch job is submitted and it will use a GPU, make sure no interactive sessions are using the GPU or the batch job will wait for the interactive job to finish.

Batch jobs

You **must** set the project for this job to *medaihack*

CPU-only jobs, skip these 2 lines

- Please refer to our documentation on [batch](#) (non-interactive) jobs.
- Here is a sample GPU job script:
 - A copy of this is in `/projectnb/medaihack/startup/sample.qsub`
 - Anything the job script prints will go to a file named after the job name and the job number: *teamN_training.o123456*
- Submit this job (saved to file *train.qsub*):
 - **qsub train.qsub**

```
#!/bin/bash -l

# You *MUST* include this
#$ -P medaihack

# Request your GPU
#$ -l gpus=1
#$ -l gpu_c=8.9
#$ -l gpu_type=L40S
# For GPU jobs, request 8 cores. CPU jobs can
# be 1,2,4,8,16,28,32
#$ -pe omp 8

# Send yourself an email when it's done
# or is cancelled:
#$ -m ea
# Add your team to the emails. Don't forget
# to add YOUR email to this list.
#$ -M myself@bu.edu,member1@bu.edu,member2@bu.edu

# Give the job a name
#$ -N teamN_training

# Join stdout and stderr into one file
#$ -j y

module load medaihack/spring-2026
module load python3/3.12.4

# Activate your venv
source /projectnb/medaihack/YOUR_TEAM/venv_name

# Do stuff. As many commands as you want.
python my_training.py ...args....
```

Links to RCS Documentation

- [OnDemand](#) documentation
- [Batch](#) (non-interactive) Jobs
- [Linux guide](#) for SCC users.
- Linux usage [quick reference](#).
- Quick reference for [SCC commands](#).